

Lecture 11 Worksheet - Dialect Conversion

목표: NPU legal op set 을 정의하고 unsupported op lowering pass skeleton 을 작성합니다.

Exercise A - NPU Legal Op Set 작성

Op	Legal?	Shape guard	DType guard	Layout guard	Lowering strategy	Diagnostic
matmul						
conv2d						
elementwise						
reduction						
gather						
gelu/erf						
softmax						

Exercise B - Dynamic Legality Predicate

아래 pseudo code 의 TODO 를 채우세요.

```
target.addDynamicallyLegalOp<npu_graph::MatMulOp>([](npu_graph::MatMulOp op) {  
    // TODO 1: static shape check  
    // TODO 2: K dimension is multiple of MMA_K  
    // TODO 3: element/accumulator dtype policy  
    // TODO 4: layout and epilogue policy  
    return true;  
});
```

Exercise C - Unsupported Op Lowering Matrix

Unsupported op	Root cause	Direct/decompose/ approx/fallback	Error budget	Required tests

Exercise D - FileCheck Test 작성

```
// CHECK-LABEL: func.func @matmul_bias_gelu  
// CHECK-NOT: npu_graph.gelu_exact  
// CHECK: "npu_graph.matmul_epilogue"  
// CHECK-SAME: epilogue = "gelu_lut"
```

Submission Checklist

- legal op set YAML 1 개
- ConversionTarget skeleton 1 개
- ConversionPattern skeleton 1 개
- positive/negative FileCheck test 각 1 개
- unsupported op lowering matrix 1 개